For use by the Project lecturer	Approved		Revision required	

Feedback

For use by the Project module lecturer only

To be completed by the student

PROJECT PROPOSAL 2025

Project no	JPJ1	Revision no	0

Title	Surname	Initials	Student no	Study leader (title, initials, surname)
Mr	**Chunder**	**H**	22659511	Prof. J.P. Jacobs

Project title (this is the title on the project concept note)

App for detecting music instruments in polyphonic recordings

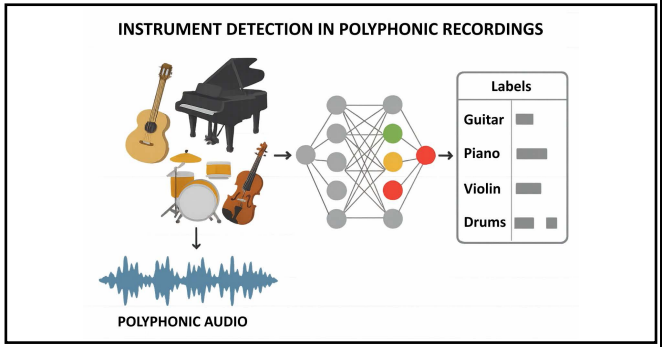
Language editor details	Language editor signature
Mr. M.G. Le Roux	
Student declaration I understand what plagiarism is and that I have to complete my project on my own.	**Study leader declaration** This is a clear and unambiguous description of what is required in this project. Approved for submission (Yes/No)
Student signature	Study leader signature and date 2025-04-22

1. Project description.

What is the problem to be solved with your project? Describe in non-technical language what your project is about. What does your system have to do? Refer to Figure 1 that you inserted on the right.

Within the field of music information retrieval (MIR), a fundamental challenge is the task of identification of instruments within polyphonic recordings. Whilst a musical expert may be able to correctly identify which instruments are present at a given moment within a mix, this remains a challenge for individuals with limited musical training, or for experts having to work with several hours worth of recordings. Knowledge of the instrument presence at a given moment within a polyphonic mix has several practical applications, including: synchronization of lighting effects during multimedia performances or identification of an instrument's entry time when editing a recorded mix. The project aims to develop an application which can listen to, and identify the constituent instruments within a polyphonic mix. The application should be able to answer queries such as: "Is a piano present in the mix? If so, what time does the piano enter the mix?" The genre of the polyphonic recordings is limited to classical music. The polyphonic recordings are limited to duet, trio, quartet and quintet classical ensembles. Identification is limited to eleven classical musical instruments: violin, viola, cello, contrabass, clarinet, flute, bassoon, French horn, trumpet and piano.

Figure 1. Insert an image below that captures the idea with your project. Explain in the text on the left.



2. Technical challenges in this project

Describe the technical challenges that are *beyond* those encountered up to the end of third year and in other final year modules.

2.1 Primary *design* challenges

Which aspects of the design of the system do you expect to be the most challenging?

• The complexity of the spectral content on the input polyphonic signal will reflect significant variety with respect to the instruments which are present in the mix. This a poses significant challenge as the timbre of each instrument may vary depending on playing style and the added factor that it is not guaranteed that all instruments will be presented at the same loudness in the mix.
• The application should function reliably across different recordings and acoustic environments, including untreated environments which may introduce interference such as background noise and reverberation.
• Accurately representing the spectral content of the polyphonic signal such that timbre features for each instrument may either be hand-crafted or learned.

2.2 Primary *implementation* challenges

Which aspects of the implementation do you expect to be the most challenging?

• Porting the application to an embedded platform will require optimizing the software such that algorithms run efficiently within the hardware limits of the embedded platform.
• Achieving real-time processing on the embedded platform will be difficult; balancing latency and performance with hardware limits may necessitate trade-offs in model complexity.
• Datasets of polyphonic classical music may not posses high-quality data annotations or equal distribution of instrument presence.

Any processing will be done on an embedded platform, and not on a PC platform. Confirmed by having ticked this box. ☑

Library functions will be used for hardware interfacing only. Confirmed by having ticked this box. ☑

3. Functional analysis

3.1 Functional description

Describe the design in terms of system functions as shown on the functional block diagram in section 3.2. This description should be in *narrative format*. **DO NOT** use a bullet list.

The instrument detection application will have 3 components: signal processing (FU 2.1 - FU 2.2), data preparation (FU 3.1 - FU 3.4) and instrument inference (FU 4.1 - FU 4.2). The application will take 2 inputs (FU 1.1 - FU 1.2) and produce an output (FU 5). The user will play a polyphonic mix (FU 1.1) of duration 15 - 30 seconds, which will be an analog audio signal produced from a speaker, and will specify a query (FU 1.2) relating to the presence of an instrument within the mix. The analog signal is acquired (FU 2.1) by an external diaphragm condenser microphone, serving as a transducer, which is connected to the embedded platform. Thereafter, analog to digital conversion is applied (FU 2.2) to produce a digitized signal. Once digitized, the signal is segmented into smaller frames (FU 3.1), allowing the application a means of temporal resolution used to capture the local instrument features with their evolution over time. Each frame is transformed into the frequency-time domain via Frequency-Time Analysis (FU 3.2), allowing for either the raw frequency-time domain transformation or acoustic features hand-crafted from the frequency-time domain transformation to form the acoustic representation (FU 3.3) of the frame. The acoustic representation is formatted to an appropriate data structure corresponding to the input dimensions of the multi-label classifier (FU 3.4). This data structure is parsed as input to a multi-label classifier (FU 4.1) to perform instrument activity detection (IAD). The output of this classifier will be a vector of probabilities, relating to the likelihood of each target instrument. Decision logic (FU 4.2) is applied to determine binary activity (presence) of the queried instrument within the frame. After all frames have been parsed to the classifier, a response to the user query (FU 1.2) is determined based on the IAD output of the audio frames. This response is then displayed on a visual output (FU 5), formatted to display the instrument name and onset time in the event the queried instrument is present or to display that the queried instrument is not present at any point in the polyphonic mix.

3.2 Functional block diagram (this should not be a flow diagram)

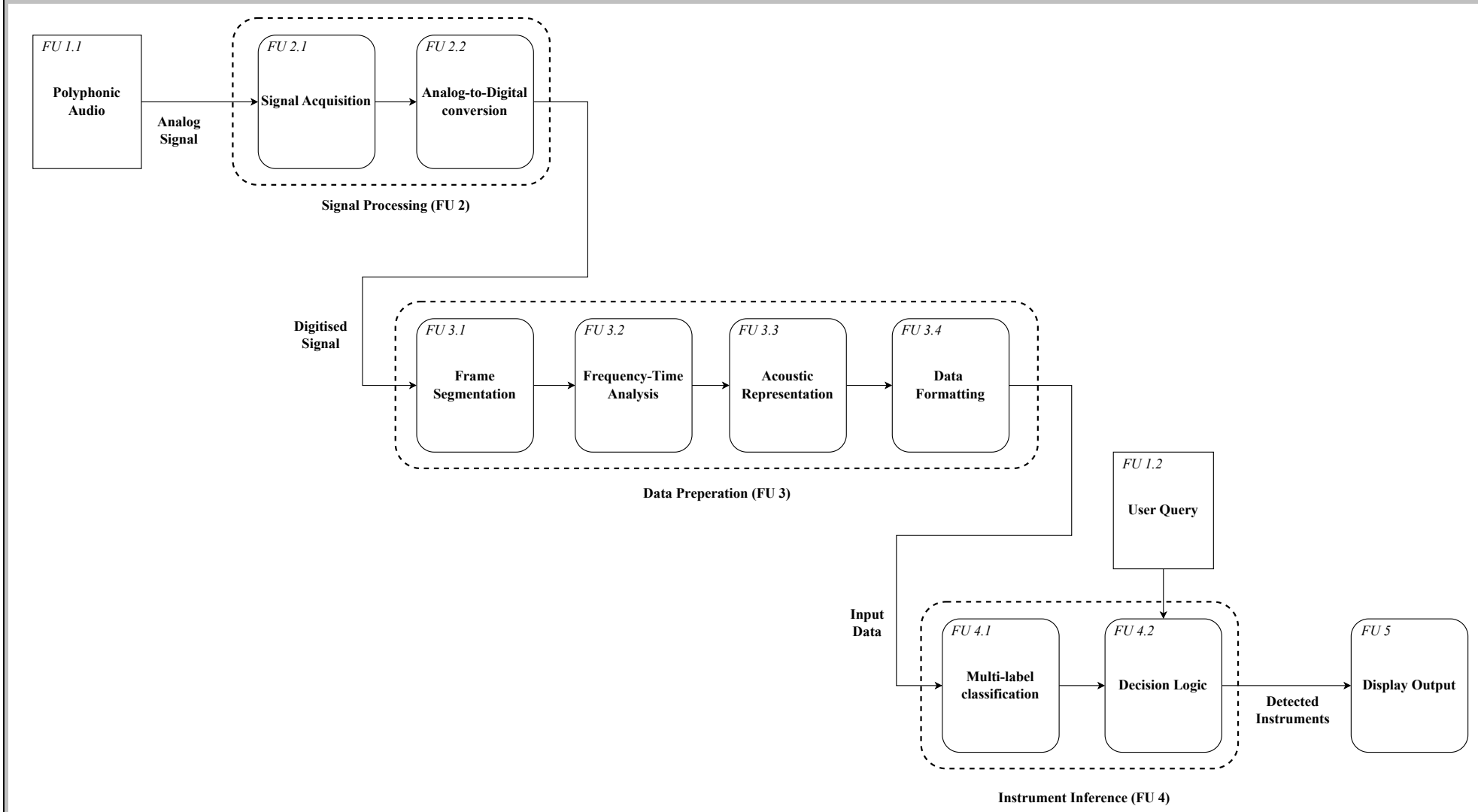


Figure 1: Functional block diagram of instrument detection application.

4. System requirements and specifications

These are the core requirements of the system or product (the mission-critical requirements) in table format **IN ORDER OF IMPORTANCE**. Requirement 1 is the most fundamental requirement.

	Requirement 1: the fundamental functional and performance requirement of your project	**Requirement 2** (Second in order of importance)	**Requirement 3** (Third in order of importance)
1. Core mission requirements of the system or product. Focus on requirements that are **core** to solving the engineering problem. These will reflect the solution to the problem.	The application must accurately and reliably detect the presence and onset time of each specified classical instrument if present within the recording.	The application must not falsely detect the presence of each specified classical instrument within the recording.	The application must correctly identify the presence and onset time of each specified classical instrument if present within the recording.
2. What is the target specification (in *measurable* terms) to be met in order to achieve the requirement in 1. above?	A minimum averaged F1 score of 0.75 across all instruments must be achieved, using a temporal collar of 100 ms applied to ground truth instrument events.	A minimum averaged precision score of 0.77 across all instruments must be achieved.	A minimum averaged recall score of 0.73 across all instruments must be achieved, using a temporal collar of 100 ms applied to ground truth instrument events.
3. Motivation: Defend the specific target specification, i.e. the value that you selected. I.e., *why* will meeting the specification given in point 2 above *solve the problem*?	The F1 score provides a balanced measure of precision and recall. An F1 score of 0.75 represents a high level of performance given 11 classification labels, based on relevant literature.	The precision score provides a measure of accuracy of positive predictions. The detection of a false positive carries more impact in real-world applications of IAD; it is difficult for a theatre performer to improvise to a false cue.	The recall score provides a measure of the model's ability to identify positive instances. The detection of a false negative carries less impact in real-world applications of IAD; it is easier for a theatre performer to improvise to a missed cue.
4. How will you demonstrate at the examination that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will *prove* these were met.	A test set of classical polyphonic recordings which have ground truth annotations will be processed. The F1 score per instrument will be calculated, and the average must equal or exceed 0.75.	A test set of classical polyphonic recordings which have ground truth annotations will be processed. The precision score per instrument will be calculated, and the average must equal or exceed 0.77.	A test set of classical polyphonic recordings which have ground truth annotations will be processed. The recall score per instrument will be calculated, and the average must equal or exceed 0.73.
5. Your own design contribution: what are the aspects that *you will design and implement yourself* to meet the requirement in point 2? If none, *remove this requirement.*	Feature representation methods to either engineer or learn timbre features, multi-label classification model, the forward pass, optimizer, learning rate scheduler, and loss function implemented from first principles.	Feature representation methods to either engineer or learn timbre features, multi-label classification model, the forward pass, optimizer, learning rate scheduler, and loss function implemented from first principles.	Feature representation methods to either engineer or learn timbre features, multi-label classification model, the forward pass, optimizer, learning rate scheduler, and loss function implemented from first principles.
6. What are the aspects to be taken off the shelf to meet this requirement? If none, indicate "none". Clearly specify for what tasks **library functions** will be used (if relevant to the project).	User interface Back propagation function Fast Fourier Transform Microphone input stream	User interface Back propagation function Fast Fourier Transform Microphone input stream	User interface Back propagation function Fast Fourier Transform Microphone input stream

System requirements and specifications page 2

	Requirement 4 (Fourth in order of importance)	**Requirement 5**	**Requirement 6**
1. Core mission requirements of the system or product. Focus on requirements that are core to solving the engineering problem. These will reflect the solution to the problem.	The application must process the input polyphonic mix and provide instrument detection results within a reasonable wait time.		
2. What is the target specification (in *measurable* terms) to be met in order to achieve the requirement in 1. above?	The total processing time of the polyphonic mix, from when the signal is acquired to when the results are displayed must not exceed 2x the duration of the polyphonic mix.		
3. Motivation: Defend the specific target specification, i.e. the value that you selected. I.e., *why* will meeting the specification given in point 2 above solve the problem?	A processing time factor of 2x provides a compromise between allowing compute time required for inference of each audio frame and the overall wait time for complete analysis of the polyphonic mix.		
4. How will you demonstrate at the examination that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will *prove* these were met.	A test set of classical polyphonic recordings will be processed with the time measured from the start of audio processing to the display of detection results. This measured time must not exceed 2x the duration for each recording.		
5. Your own design contribution: what are the aspects that *you will design and implement yourself* to meet the requirement in point 2? If none, *remove this requirement.*	Memory management algorithms, utilizing CUDA framework. Feature representation methods, multi-label classification model and model methods adopting computational efficiency principles.		
6. What are the aspects to be taken off the shelf to meet this requirement? If none, indicate "none". Clearly specify for what tasks **library functions** will be used (if relevant to the project).	GPU accelerated data structures (CUDA framework) Memory management and pointer organization Math		

System requirements and specifications page 3

	Requirement 7 (Seventh in the order of importance)	**Requirement 8**	**Requirement 9**
1. Core mission requirements of the system or product. Focus on requirements that are core to solving the engineering problem. These will reflect the solution to the problem.			
2. What is the target specification (in *measurable* terms) to be met in order to achieve the requirement in 1. above?			
3. Motivation: Defend the specific target specification, i.e. the value that you selected. I.e., *why* will meeting the specification given in point 2 above solve the problem?			
4. **How will you demonstrate at the examination** that this requirement and specification (points 1 and 2 above) have been met? Be explicit about how you will *prove* these were met.			
5. Your own design contribution: what are the aspects that *you will design and implement yourself* to meet the requirement in point 2? If none, *remove this requirement.*			
6. What are the aspects to be taken off the shelf to meet this requirement? If none, indicate "none". Clearly specify for what tasks l**ibrary functions** will be used (if relevant to the project).			

5. Field condition requirements

These are the ACTUAL, REAL WORLD CONDITIONS under which your project has to work *and will be demonstrated.*

	Real world field condition 1 (required to include)	**Real world field condition 2** (optional to include)	**Real world field condition 3** (optional to include)
Field condition requirement. In which field conditions does the system have to operate? E.g., "lab conditions" or "outside in uncontrolled conditions". Describe the one, two or three most important field conditions.	Environments which have not been acoustically treated. The polyphonic audio may be subject to mild ambient noise and reverberation present in the environment.		

6. Student tasks

6.1 Design and implementation tasks

List your primary design and implementation tasks in bullet list format (5-10 bullets). These are *NOT* product requirements, but *YOUR* tasks.

- Design and train a multi-label classifier for instruments in polyphonic recordings.
- Implement the model architecture, forward pass, loss function, optimizer and scheduler algorithms from first principles.
- Implement frequency-time analysis techniques for input representation from first principles.
- Optimization of the application software for deployment on an embedded platform.
- Design and implementation of an audio capture pipeline, including analog to digital conversion, from first principles.
- Evaluation and analysis of model performance on testing dataset.

6.2 New knowledge to be acquired

Describe what the theoretical foundation to the project is, and which *new* knowledge you will acquire (*beyond* that covered in any other undergraduate modules).

The student will have to understand theory relating to feature representation methods, such that spectral and temporal features may be hand-crafted or learned.
The student will have to understand theory relating to deep learning methods appropriate for MIR applications.
The student will have to understand theory relating to design principles of deep learning models.
The student will have to understand theory relating to statistical methods for evaluation of deep learning models and performance.
The student will have to develop an understanding of western classical music, specifically music notation.
The student will have to develop an understanding of the acoustic properties of classical musical instrument, specifically timbre.